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DUNMAN HIGH SCHOOL

Preliminary Examination

Year 6

H1 CHEMISTRY

8872/01

Paper 1 Multiple Choice

27 September 2013

50 minutes

Additional Materials: Optical Mark Sheet
 Data Booklet

INSTRUCTIONS TO CANDIDATES

- 1 Write your **name**, **index number** and **class** on this question paper.
- 2 There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.
- 3 Each correct answer will score one mark. A mark will not be deducted for wrong answer.
- 4 Any rough working should be done in this booklet.
- 5 You may use a calculator.

Section A

For each question, there are four possible answers **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 The surface area of a fine powder can be determined by measuring the volume of nitrogen gas required to form a layer one molecule thick on its surface. It is then assumed that the area of one nitrogen molecule is $Y \text{ m}^2$.

There are 6×10^{23} molecules in 1 mol of nitrogen and the molar volume of all gas at room temperature and pressure is 24000 cm^3 .

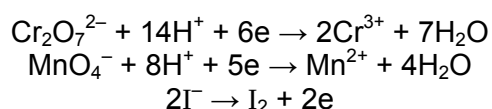
A sample of powdered titanium(IV) oxide requires $Z \text{ cm}^3$ of nitrogen to form a layer one molecule thick at room temperature and pressure.

What is the surface area of this sample in m^2 ?

- A** $(24000 / Z) \times 6 \times 10^{23} \times Y$
B $(Z / 24000) \times 6 \times 10^{23} \times Y$
C $(24000 / Y) \times 6 \times 10^{23} \times (1/Y)$
D $Z \times 24000 \times Y / (6 \times 10^{23})$
- 2 In an experiment, 10 cm^3 of an organic compound in the gaseous state was sparked with an excess of oxygen. 20 cm^3 of carbon dioxide and 5 cm^3 of nitrogen were obtained among the products. All gas volumes were measured at the same temperature and pressure.

Which of the following compounds would fit these data?

- A** $\text{C}_2\text{H}_7\text{N}$ **B** $\text{C}_2\text{H}_6\text{N}_2$ **C** $\text{C}_6\text{H}_5\text{NO}_2$ **D** $\text{C}_3\text{H}_5\text{N}$
- 3 Acidified potassium dichromate(VI) and acidified potassium manganate(VII) each liberate iodine from aqueous potassium iodide.



25.0 cm^3 of acidified $\text{Na}_2\text{Cr}_2\text{O}_7$ liberates the same quantity of iodine from a solution of potassium iodide as 25.0 cm^3 of $0.030 \text{ mol dm}^{-3}$ KMnO_4 . What is the concentration of the $\text{Na}_2\text{Cr}_2\text{O}_7$ solution in mol dm^{-3} ?

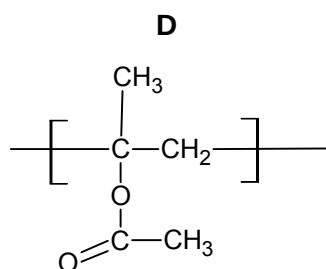
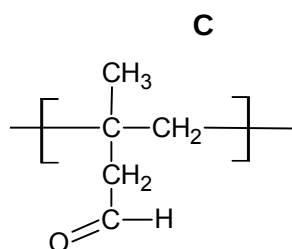
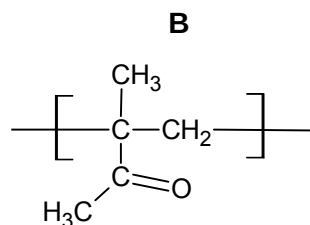
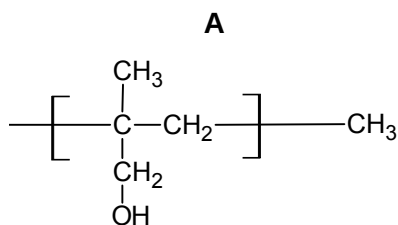
- A** 0.015 **B** 0.025 **C** 0.030 **D** 0.050
- 4 Which one of the following is **not** a redox reaction?
- A** $\text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O} \rightarrow \text{SO}_4^{2-} + 4\text{H}^+ + 2\text{Br}^-$
B $6\text{NaOH} + 3\text{Br}_2 \rightarrow 5\text{NaBr} + \text{NaBrO}_3 + 3\text{H}_2\text{O}$
C $2\text{CuSO}_4 + 4\text{KI} \rightarrow 2\text{CuI} + 2\text{K}_2\text{SO}_4 + \text{I}_2$
D $2\text{CrO}_4^{2-} + 2\text{H}^+ \rightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$

- 5 The number of neutrons and electrons of the following particles are given below:

Particle	No. of neutrons	No. of electrons
P	15	11
Q^{2-}	17	11
R^+	15	10
S^-	17	12
T^+	16	13

Which atom is an isotope of particle P?

- A Q B R C S D T
- 6 The successive ionisation energies, in kJ mol^{-1} , of an element Z are given below.
 1010 2265 3662 4998 7002 8992 28698 32215
- Which one of the following would best represent the formula of the ionic compound formed by Z?
- A ZCl
 B ZO_2
 C CaZ_2
 D Na_2Z
- 7 Which one of the following ions will be deflected to the greatest extent in an electric field?
- A ${}^6\text{Li}^+$ B ${}^{27}\text{Al}^{3+}$ C ${}^{16}\text{O}^{2-}$ D ${}^{14}\text{N}^{3-}$
- 8 The following polymers could be used for contact lenses. A contact lens has to absorb water so that it can fit comfortably in the eye.
- Which polymer will absorb water most easily?



- 9 The table shows the enthalpy change of neutralisation per mole of water formed, ΔH , for various acids and bases.

Acid	Base	$\Delta H / \text{kJ mol}^{-1}$
Hydrochloric acid	Sodium hydroxide	-57.0
P	Sodium hydroxide	-54.0
Hydrochloric acid	Q	-52.0
Nitric acid	R	-57.0

What are **P**, **Q** and **R**?

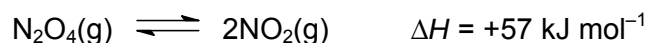
	P	Q	R
A	Ethanoic acid	Ammonia	Potassium hydroxide
B	Ethanoic acid	Sodium hydroxide	Ammonia
C	Sulfuric acid	Ammonia	Potassium hydroxide
D	Sulfuric acid	Sodium hydroxide	ammonia

- 10 The radius and ionic charge of each of six ions are shown in the table below.

<i>Ion</i>	J^+	L^+	M^{2+}	X^-	Y^-	Z^{2-}
<i>Radius/nm</i>	0.14	0.18	0.15	0.14	0.18	0.15

The ionic solids JX, LY and MZ are of the same lattice type. What is the correct order of the lattice energies of JX, LY and MZ, placing the one with the highest numerical value first?

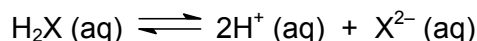
- A** JX > LY > MZ
B LY > MZ > JX
C MZ > JX > LY
D MZ > LY > JX
- 11 The dissociation of dinitrogen tetroxide into nitrogen dioxide is represented by the equation below.



Which of the following statement correctly explains how the volume of the mixture will be affected if the temperature of an equilibrium mixture of the gases is increased at constant pressure?

- A** The volume will increase, only because of a shift of equilibrium towards the right.
B The volume will increase, both because of a shift of equilibrium towards the right and also because of thermal expansion.
C The volume will stay the same, because any thermal expansion could be exactly counteracted by a shift of equilibrium towards the left.
D The volume will decrease, because of a shift of equilibrium towards the left would more than counteract any thermal expansion.

- 12 The equilibrium equation for the dissociation of a weak acid, H_2X , may be written as follows.



What is the expression for the acid dissociation constant, K_a ?

A
$$\frac{[\text{2H}^+][\text{X}^{2-}]}{[\text{H}_2\text{X}]}$$

C
$$\frac{[\text{H}^+][\text{X}^{2-}]}{[\text{H}_2\text{X}]}$$

B
$$\frac{[\text{H}^+]^2[\text{X}^{2-}]}{[\text{H}_2\text{X}]}$$

D
$$\frac{[\text{2H}^+]^2[\text{X}^{2-}]}{[\text{H}_2\text{X}]}$$

- 13 An enzyme, found in the stomach, operates at maximum efficiency when in an aqueous solution buffered at pH 5.

Which combination of substances, when dissolved in 10 dm^3 of water, would give the necessary buffer solution?

- A** 2 mol of $\text{CH}_3\text{CO}_2\text{H}$ and 1 mol of NaOH
- B** 2 mol of NaOH and 1 mol of $\text{CH}_3\text{CO}_2\text{H}$
- C** 1 mol of HCl and 1 mol of $\text{CH}_3\text{CO}_2\text{Na}$
- D** 2 mol of NH_3 and 1 mol of $\text{CH}_3\text{CO}_2\text{NH}_4$

- 14 A chemist discovers that 5 g of an iron(II) salt, when used in a certain reaction at a certain temperature, causes the reaction to speed up. When he tries another 5 g of the same salt in another reaction at the same temperature, the reaction rate is still the same even after a long period of time. What is the most likely reaction for this?

- A** The chemist did not stir the reaction mixture the second time.
- B** The temperature is not high enough for reaction rate to increase.
- C** The iron(II) salt has a specific nature as a catalyst.
- D** Too little of the iron(II) salt is used.

- 15 The half-life of a radioactive isotope **T** is twice that of another radioactive isotope **U**. In a sample of rock, it is found that the number of atoms of **T** is 4 times that of **U**. What will be the ratio of the number of atoms of **T** to the number of atoms of **U** in the rock after two half-lives of **T**?

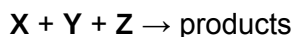
A $\frac{1}{16}$

B $\frac{1}{4}$

C $\frac{1}{2}$

D 1

- 16 The stoichiometry chemical equation for a hypothetical reaction may be represented as

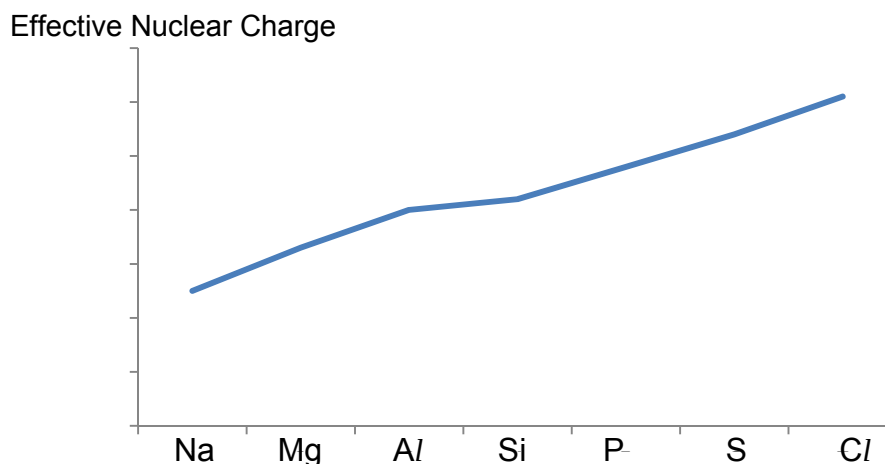


The reaction was studied in a series of four experiments in which the initial rate of the reaction was measured as a function of the concentration of each reactant with the following results:

Experiment	[X] / mol dm ⁻³	[Y] / mol dm ⁻³	[Z] / mol dm ⁻³	Initial rate / mol dm ⁻³ s ⁻¹
1	0.3	0.1	0.01	6×10^{-6}
2	0.3	0.2	0.01	12×10^{-6}
3	0.4	0.2	0.01	16×10^{-6}
4	0.4	0.1	0.02	8×10^{-6}

Which of the following shows the correct rate equation?

- A Rate = $k[\text{X}][\text{Y}]$
 - B Rate = $k[\text{X}][\text{Y}][\text{Z}]$
 - C Rate = $k[\text{Y}][\text{Z}]$
 - D Rate = $k[\text{X}]^2[\text{Z}]$
- 17 The following shows a graph of effective nuclear charge against proton number for elements in Period 3 of the Periodic Table, from sodium to chlorine.



Which of the following trends **cannot** be inferred from the graph above?

- A Atomic radii decrease across the Period.
- B Electron affinity increase across the Period.
- C Electronegativity increase across the Period.
- D Electrical conductivity decrease across the Period.

- 18 **F**, **G** and **H** are consecutive elements in Period 3. Element **H** has an oxide that reacts with hot concentrated NaOH, and the first ionisation energy of element **G** is lower than **F**.

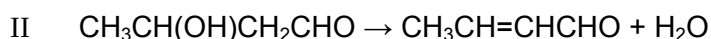
What is the identity of elements **F**, **G** and **H**?

	F	G	H
A	Mg	Al	Si
B	Na	Mg	Al
C	Al	Si	P
D	Si	P	S

- 19 Which of the following alcohols produce the largest number of products, including stereoisomers, when reacted with excess concentrated H₂SO₄ at 170 °C?

- A** CH₃CH₂CH₂CH₂CH₂OH
B CH₃CH₂CH₂CH(OH)CH₃
C CH₃CH₂CH(OH)CH₂CH₃
D CH₃CH(CH₃)CH(OH)CH₃

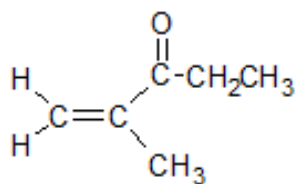
- 20 The Russian composer Borodin was also a research chemist who discovered a reaction in which two ethanol molecules combine to form a compound as shown in reaction I. The product forms another compound on heating in reaction II.



Which of the following best describes reactions I and II?

	I	II
A	Addition	Elimination
B	Addition	Reduction
C	Elimination	Reduction
D	Substitution	Elimination

- 21 Molecule **Z** is a derivative of a compound used to make the hard outer covering of golf balls.

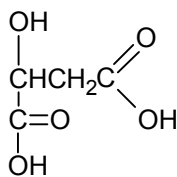


Which of the following is **not** true about **Z**?

- A It is insoluble in water.
 - B It can undergo a condensation reaction to give an orange compound.
 - C It can react with HBr at room temperature and pressure.
 - D It gives a secondary alcohol as a major product when reacting with steam and phosphoric acid.
- 22 A sample of 1,3-dimethylbenzene ($M_r = 109$) is oxidised by acidified potassium manganate(VII) to the corresponding dicarboxylic acid.

What is the mass of the organic product formed from 1.00 g of 1,3-dimethylbenzene?

- A 1.15 g
 - B 1.28 g
 - C 1.57 g
 - D 1.60 g
- 23 Malic acid is found in apples.



Malic acid

Which substance will react with all three -OH groups present in the malic acid molecule?

- A Phosphorus pentachloride
- B Potassium hydroxide
- C Ethanol in the presence of concentrated sulfuric acid
- D Sodium carbonate

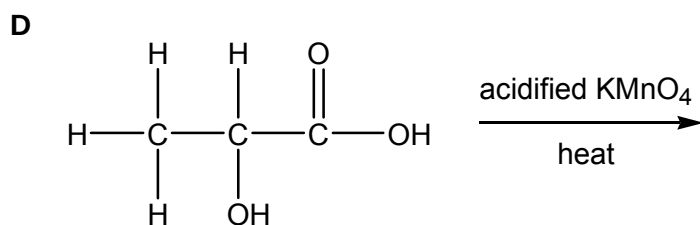
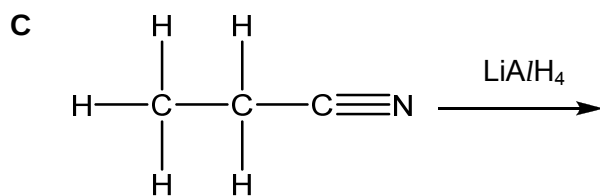
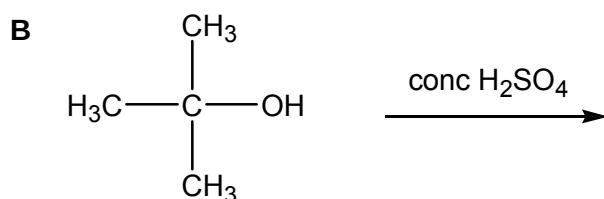
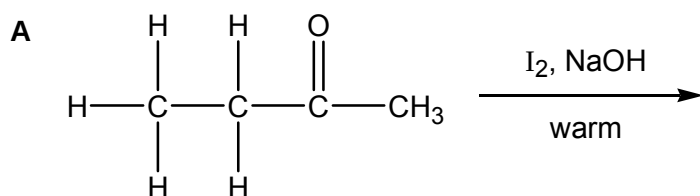
- 24 The compound $\text{C}_3\text{H}_7\text{Br}$ undergoes a sequence of reactions as follows:



What are the compounds X, Y and Z?

	X	Y	Z
A	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	$\text{CH}_3\text{CH}_2\text{CHO}$
B	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	$\text{CH}_3\text{CH}_2\text{CHO}$	$\text{CH}_3\text{CO}_2\text{H}$
C	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	$\text{CH}_3\text{CH}_2\text{CHO}$	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$
D	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	CH_3COCH_3	$\text{CH}_3\text{CO}_2\text{H}$

- 25 Which reaction yields an organic product that has **more** hydrogen atoms than the reactant?



Section B

For each question in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements which you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 26** Carbon exists as a number of allotropes, for example graphite, fullerenes and acetylenic carbon, which have the following description.

Allotrope	Description
Graphite	Each carbon atom is covalently bonded to three other carbon atoms.
Fullerene	Each carbon atom has an unhybridised p orbital with one electron, which can be delocalised over the whole molecule.
Acetylenic carbon	Chains of atoms with alternating triple and single bonds.

Which of the following deductions on the shape of the carbon atom and hybridisation of the allotrope are correct?

	Allotrope	Shape	Hybridisation
1	Fullerene	Trigonal planar	sp^2
2	Acetylenic carbon	Linear	sp
3	Graphite	Trigonal pyramidal	sp^3

- 27** Which statements are correct for the sequence of compounds below, considered from left to right?



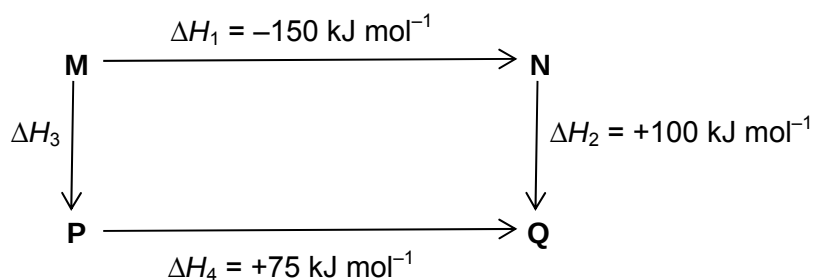
- The electronegativity difference between the elements in each compound increases.
- The formula–units of these compounds are isoelectronic (have the same number of electrons).
- The bonding becomes increasingly covalent.

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

28 The diagram below illustrates the energy changes for a set of reactions.



Which statements about the energy changes involved are correct?

- 1** The value of $\Delta H_1 + \Delta H_2$ is equal to $\Delta H_3 + \Delta H_4$.
- 2** The enthalpy change of the reaction $\text{M} \rightarrow \text{P}$ is -125 kJ mol^{-1} .
- 3** The enthalpy change of the reaction $\text{P} \rightarrow \text{N}$ is $+25 \text{ kJ mol}^{-1}$.

29 In which of the following reactions does NH_3 react as a Bronsted–Lowry base?

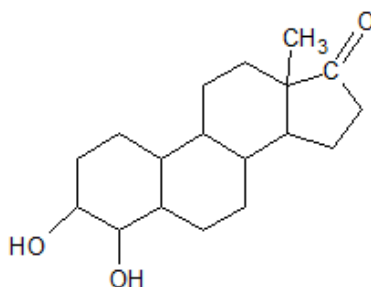
- 1** $\text{HSO}_4^- + \text{NH}_3 \rightarrow \text{SO}_4^{2-} + \text{NH}_4^+$
- 2** $2\text{NH}_3 \rightarrow \text{NH}_2^- + \text{NH}_4^+$
- 3** $\text{Ag}^+ + 2\text{NH}_3 \rightarrow [\text{Ag}(\text{NH}_3)_2]^+$

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

30



The steroid shown is an intermediate compound obtained during the synthesis of Formestane which is used in the treatment of breast cancer.

Which statement about this compound is correct?

- 1** It reacts with hydrogen cyanide in an addition reaction.
- 2** It can be oxidised by warm acidified potassium dichromate(VI) to a carboxylic acid.
- 3** It will react with Fehling's solution.

Answers

1	2	3	4	5	6	7	8	9	10
B	A	B	D	C	D	D	D	A	C
11	12	13	14	15	16	17	18	19	20
B	B	A	C	D	A	D	A	B	A
21	22	23	24	25	26	27	28	29	30
D	C	A	C	C	B	C	B	B	D